

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

QUESTIONS: 5

Total Marks:50

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Dharani Nikesh N S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dharaninikesh N S

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Answer:

IPv4 Subnet Debugging

- IP: 192.168.10.65/26
- Network: **192.168.10.64/26**
- Broadcast: **192.168.10.127**
- Host range: **192.168.10.65 – 192.168.10.126**
- IP 192.168.10.65: **Valid**
- Gateway 192.168.10.1: **Invalid for this subnet**
- Error: IP and gateway are in different /26 subnets.
- Correction: IP = 192.168.10.10, Gateway = 192.168.10.1
- Usable hosts: **62**

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer:

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B:
2001:0db8:0000:0000:0001:0000:0000:0001/64

- Network prefix for both: 2001:db8:0:0::/64
- Same subnet: **Yes**
- Prefix mismatch: **No**
- Possible error: Firewall, VLAN, Neighbor Discovery, or connectivity issue.
- Correct example:
 - A = 2001:db8:0:0::1/64
 - B = 2001:db8:0:0::2/64
- IPv6 addresses: $2^6 \cdot 4 = 18,446,744,073,709,551,616$

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Answer:

Subnet	Network	Broadcast	Usable Hosts
/26	192.168.1.0	192.168.1.63	62
/27	192.168.1.64	192.168.1.95	30
/28	192.168.1.96	192.168.1.111	14

- /26 → 62 usable
- /27 → 30 usable
- /28 → 14 usable
- Problem: Large subnets may waste addresses.
- Solution: Use **VLSM** according to department size.
- Example: 50 hosts → /26, 25 hosts → /27, 10 hosts → /28.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer:

- A prefix: 2001:db8:0:0::/64
- B prefix: 2001:db8:0:0::/64
- Same subnet: **Yes**
- Prefix mismatch: **No**
- Correct example:
 - A = 2001:db8:0:0::1/64
 - B = 2001:db8:0:0::2/64
- Available addresses: **2⁶⁴ - 4**

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Answer:

Destination: 192.168.2.100

- Matching route: 192.168.2.0/24
- Next hop: 10.0.0.3
- Destination belongs to subnet: **Yes**
- Routing error: No specific route error is shown.
- If no matching route exists → use **Default Route** 0.0.0.0/0 → 10.0.0.1
- Optimization: Verify next-hop reachability and return route.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding(2)	Identifies most errors with minor gaps(1.5)	Identifies some errors but misses key issues(1)	Unable to identify errors correctly(0)
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues(2.5)	Uses appropriate debugging methods with minor errors(2)	Limited debugging approach; requires guidance(1.5)	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms(2)	Improves time complexity with minor gaps in efficiency(1.5)	Limited improvement in time complexity(1)	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions(2)	Shows reasonable analysis with some justification(1.5)	Limited analysis; weak reasoning(1)	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation(1.5)	Code is mostly clear with minor documentat ion gaps(1)	Code lacks structure and proper documentatio n(0.5)	Poorly written code with no documentatio n